AI-Based System for Underwater Monitoring: Image Enhancement, Detection, and Depth Mapping

Archana Kotakar| Pallavi Rajendra Gawai| Priyanka Prabhakar Gujale|

Dnyaneshwari Baliram Suke| Kajal Ashok Shelke

Introduction

Monitoring underwater ecosystems is highly important for environmental protection, marine research, and pollution control. However, due to issues like poor lighting, water turbidity, and suspended particles, captured underwater images are usually unclear. Traditional methods such as sonar equipment or human divers are costly, time-consuming, and sometimes risky. To overcome these problems, this project proposes an AI-powered system that automatically improves underwater image clarity, detects relevant objects (such as fish, algae, and plastic), and estimates the depth of water zones using deep learning algorithms. The results are then presented through an interactive dashboard, allowing users to visualize, analyze, and make effective decisions for conservation and monitoring.

Objective

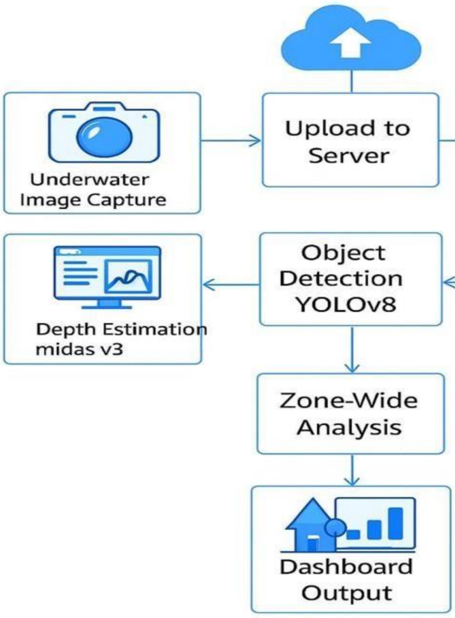
1. To design a system that captures underwater images using low-cost waterproof cameras.
2. To enhance the quality of captured images using OpenCV/ESRGAN.
3. To detect objects such as fish, corals, plastic, and waste with YOLOv7/YOLOv8.
4. To predict depth using MiDaS v3.1 (DPT-Hybrid) without expensive sonar sensors.
5. To apply K-Means clustering and Decision Tree algorithms for zone-wise analysis.
6. To build a Streamlit dashboard for visualizing enhanced images, detected objects, and depth maps.

Problem Formulation

Current underwater monitoring methods depend on manual diving, sonar sensors, or expensive specialized equipment, which are not scalable and often perform poorly in lowvisibility conditions. The major challenges include poor image visibility due to scattering and distortion, difficulty in detecting small or hidden objects, high costs of sonar-based systems, and the lack of accessible AI-driven monitoring solutions. This project aims to overcome these issues by developing a cost-effective AI-based system that enhances underwater images, accurately detects objects, and efficiently estimates depth.

Methodology

The methodology involves capturing underwater images using cameras such as GoPro or DJI Osmo. These images are preprocessed with OpenCV techniques like denoising, resizing, and normalization. To enhance clarity, deblurring, contrast adjustment, and color correction are applied. Object detection is performed using YOLOv7/YOLOv8 to identify fish, algae, plastic, and other objects, while MiDaS v3.1 (DPT-Hybrid) is used for monocular depth estimation. For zone-wise analysis, K-Means clustering classifies water regions, and Decision Tree algorithms evaluate pollution levels. Finally, all outputs, including enhanced images, detection results, and depth maps, are visualized on a Streamlit-based dashboard



Expected Outcome

The expected outcome of this system includes the generation of enhanced and clear underwater images, accurate detection of key objects like fish, algae, and plastic, and reliable depth estimation maps with zone-wise pollution details. The Streamlit dashboard provides comprehensive visual reports, aiding in informed decision-making. Overall, the system aims to deliver a scalable, cost-effective, and efficient solution that can be utilized by researchers, environmentalists, and government organizations for underwater monitoring and environmental protection.